

Adapting Before Contact: Compositional Contextual Memory for Frozen World Models

Abstract

Error-driven world-model adaptation is reactive: the transition that reveals a dynamics mismatch has already occurred. In striking, throwing, and other single-contact tasks, its feedback can arrive after the consequential action is committed. We formulate this first-contact limitation and introduce CIRA (Context-Indexed Residual Adapters), which retrieves previously learned dynamics corrections before interaction. A frozen visual world model is augmented with persistent probabilistic memories, a context posterior combining cues and dynamics evidence, and a factor-structured prior that transfers knowledge to unseen combinations. Memory updates use context probabilities fixed before the current prediction error, permitting a uniform error bound that separates estimation noise, prior mismatch, and contamination by other contexts. The analysis connects memory accuracy and context log loss to first-contact prediction and decision error, and characterizes the observation graph required to identify an unseen additive combination. The same contextual belief supports decision-aware probing and reuse of safety margins: information has zero decision value when all plausible models share an optimal action, while a per-memory calibration rule bounds long-run latent barrier failures under explicit enforcement and boundedness assumptions. A counterfactual decomposition separates improvement through memory selection from improvement through parameter learning. Controlled evaluation protocols isolate recurrence, cue reliability, compositional transfer, calibration, and probing, with the first consequential prediction as the primary endpoint. The resulting framework treats adaptation as inference over a reusable physical repertoire, with new learning reserved for uncertainty that experience has not already resolved.

1 Introduction

Learned world models support robot control by predicting the consequences of candidate actions. Visual models make this approach possible directly from images, using a learned representation in which a planner can compare alternative futures [8]. Deployment, however, changes the relationship between action and outcome. A familiar object slides differently on another table, a container carries a different load, or a replacement tool changes the robot’s effective dynamics. Online adaptation addresses these discrepancies by updating a model from the errors observed during execution. Recent approaches update world-model parameters or lightweight residual modules using transitions collected while planning [6, 7].

This strategy has a timing limitation. Consider a robot striking a puck toward a target. The appropriate strike depends on the surface and on the object’s response to impact. If the robot assumes too much friction, the puck may overshoot the target or leave the table. The prediction error reveals the mismatch, but only after the strike has been committed. Correcting the model at that point can improve the next attempt; it cannot change the impulse already delivered. Similar timing constraints arise in throwing, in a lift whose initial force can cause slipping or excessive acceleration, and in pushing sufficiently close to an edge that the initial motion is consequential.

These are *single-contact tasks* in the sense relevant here: an outcome component is determined by an action committed before feedback from that interaction can affect the decision.

For an adaptation rule driven exclusively by current-episode prediction errors, the limitation is immediate. Before any eligible error feedback has arrived, the predictor remains at its episode initialization. If that initialization is the frozen model, its first-contact prediction is exactly the frozen prediction. This observation does not depend on the optimizer, the number of parameters being adapted, or how quickly adaptation proceeds once data arrive. It identifies an information boundary: improving the consequential first prediction requires useful information available before the error it would otherwise generate.

Many deployments provide such information through recurrence. A robot may repeatedly encounter the same surfaces, payload classes, containers, and tools. Treating each return as a fresh estimation problem discards knowledge that could already be useful. In this setting, adaptation includes a second question alongside parameter estimation: *which previously encountered physical context is active now?* Maintaining separate dynamics memories makes this question operational. A robot can preserve what it learned about one condition while operating in another, then retrieve that knowledge when the earlier condition returns. Persistent mixtures of dynamics models already establish a precedent for this form of reuse [5]; the first-contact setting asks when such knowledge can influence an action before new dynamics evidence is available.

Contextual cues provide one possible answer. A surface may look like felt or acrylic, a container may identify a recurring payload class, and a visible tool may indicate which actuator response to expect. These cues are imperfect evidence. Identical appearances can conceal different physics, and familiar physics can acquire a new appearance. We therefore use cues to infer a distribution over robot-discovered dynamics memories. The robot learns what a cue predicts through interaction, and subsequent dynamics evidence can confirm or overturn the resulting belief. The relevant operation is retrieval of learned physical experience, rather than an assumption that appearance uniquely determines physics. A truly novel condition with no transferable structure remains uncertain before it is encountered.

Recurrence also has compositional structure. A robot may have experienced surface B with a light load and a heavy load on surface A, yet never encountered surface B with the heavy load. Independent memories indexed only by the complete condition cannot directly reuse these experiences as a new combination. A shared prior over factor effects can transfer their contributions, provided the effects are sufficiently separable and previous experience identifies the required relations. This leads to a stronger test than rapid adaptation after a change: prediction at the first contact of the first-ever occurrence of a held-out combination. Independent interventions do not guarantee additive dynamics, so the prior also represents combination-specific interactions and the uncertainty they leave unresolved.

We introduce CIRA (Context-Indexed Residual Adapters), a persistent library of compact probabilistic corrections around a frozen visual world model. Before contact, a context posterior combines deployment history with the current cue to retrieve a distribution over corrections. After contact, dynamics likelihoods revise this belief, and sustained evidence of unexplained dynamics supports a new memory. A shared Gaussian prior organizes complete-context memories by the physical factors they have in common. This gives an unseen combination a predictive distribution before its first interaction, including uncertainty about its unobserved interaction component.

Learning those memories introduces a second role for context inference. A model that stores an error according to how well that same error matches a memory can systematically select its observation noise. CIRA therefore weights each update by the context probabilities available before observing the current residual. We derive a uniform memory-error bound for this predictable update. Its terms distinguish observation noise, mismatch in the transferred prior, and contamination by

incorrectly assigned contexts. The context filter’s log loss controls the accumulated contamination and also bounds first-contact selection error. Under Lipschitz dynamics and costs, these errors translate into a gap relative to the best candidate committed action. The analysis thus links the quality of a contextual repertoire to the decisions it can support.

Contextual uncertainty also determines whether additional interaction is useful. In a separate probe-enabled protocol, the robot can gather evidence before committing to the strike. A probe is selected by its expected effect on the decision, accounting for its cost. A value-of-information result shows why identifying the physical condition need not help: if all plausible models share an optimal action, informative observations have zero decision value. The same posterior supports a safety filter with per-memory calibration, so a familiar condition can recover a previous uncertainty margin. A long-run bound concerns failures of the latent barrier condition under explicit boundedness and enforcement assumptions; it does not establish physical safety from context recognition alone.

The central empirical question is whether this architecture changes the mechanism of adaptation. A faster loss decrease alone cannot distinguish retrieving an existing model from fitting a new one. A counterfactual replay decomposition separates changes in memory selection from changes in memory parameters and transferred prior centres. A complementary intervention freezes learned memories during later returns. Controlled changes to cue informativeness, recurrence, factor combinations, and intervening experience isolate when persistent context inference helps, while calibration and probing experiments test the additional predictions of the analysis.

The resulting perspective changes the unit of adaptation. Deployment becomes inference over a growing repertoire of physical contexts, with parameter learning used to acquire or refine the memories in that repertoire. Recurrence then allows previously learned dynamics to influence the first action of a new episode, while composition allows familiar physical attributes to contribute predictions in unfamiliar combinations. This paper develops four contributions:

- **First-contact adaptation and control.** We formalize the timing limitation of error-driven episode resets and connect contextual prediction accuracy to the cost of the committed action.
- **Persistent compositional memory.** We introduce cue-guided residual retrieval with a shared factor prior and characterize when observed combinations identify a new additive combination, including the uncertainty floor left by independent interactions.
- **Learning and acting under context uncertainty.** We bound memory error under predictable context-weighted updates, characterize decision-relevant probing, and establish a conditional long-run bound for memory-specific latent safety calibration.
- **Mechanisms of adaptation.** We separate recall from parameter learning and define controlled evaluations of first contact, recurrence, composition, interference, calibration, and information gathering.

2 Problem formulation and preliminaries

2.1 Frozen prediction under recurring physical contexts

Let $o_{e,t}$ and $a_{e,t}$ denote the observation and action at step t of episode $e \in \{1, \dots, N\}$. A frozen encoder and predictor produce

$$z_{e,t} = E(o_{e,t}) \in \mathbb{R}^d, \quad \hat{z}_{e,t+1}^{\text{base}} = f(z_{e,t}, a_{e,t}). \quad (1)$$

Each episode has an unobserved physical context c_e that is fixed within the core task. Its dynamics satisfy

$$z_{e,t+1} = f(z_{e,t}, a_{e,t}) + \mu_{c_e}(z_{e,t}, a_{e,t}) + \epsilon_{e,t}, \quad \mathbb{E}[\epsilon_{e,t} \mid z_{e,t}, a_{e,t}, c_e] = 0. \quad (2)$$

Here μ_c is the conditional mean discrepancy from the frozen predictor, and the noise may depend on state and action. Context labels and physical parameter values are unavailable to the robot. Episode boundaries are observed, and data and memories may persist across them. Let $D_{<e}$ denote the observations, cues, actions, and transitions available before episode e .

The robot receives a contextual cue y_e before its first informative interaction. Write τ_e for the decision step at which the corresponding action is committed; its context-dependent transition is observed afterward. The cue includes the context-relevant information available in the initial scene. For pre-contact motion history H_e^{pre} , the core setting assumes

$$I(c_e; H_e^{\text{pre}} \mid y_e, D_{<e}) = 0. \quad (3)$$

Thus moving to the interaction pose provides no additional identification of the current physics. This is a restriction on the information available in the task, not a claim that all robotic motion is uninformative.

2.2 The first-contact limitation

Definition 1 (Single-contact task). *A task is single-contact with respect to an outcome component J_e^{first} if an action a_e^{first} , or a fixed action segment, is committed at τ_e before feedback from its context-dependent transition is available, and later actions cannot change J_e^{first} . The outcome component must be specified in advance and carry task cost or a safety consequence. A strict single-contact task has its entire task outcome determined this way.*

For puck striking, J_e^{first} can be the landing or stopping error after a single committed strike. For lifting, it can be peak acceleration during the initial force application; the later placement need not be irreversible. The definition concerns the commitment and feedback times, rather than literally counting all geometric contacts. A diagnostic nudge before the strike changes the information available and is excluded from the core protocol.

Observation 1 (First-contact initialization). *Consider an adaptive predictor whose episode state $S_{e,t}$ changes only upon receipt of eligible current-episode prediction-error feedback, and whose prediction has no other adaptation path. If no such feedback is available before a_e^{first} is committed, then $S_{e,\tau_e^-} = S_{e,0}$. Consequently, its prediction for every given (z, a) at that time equals its episode-initialized prediction. If the state is reset to the frozen predictor at each episode boundary, then*

$$\hat{z}_{e,\tau_e+1}^{\text{adapt}}(z, a) = f(z, a). \quad (4)$$

This identity is about prediction before feedback. Equality of task outcomes additionally requires the same planning rule, inputs, and randomization. The observation does not cover persistent initialization, cue-driven retrieval, auxiliary pre-contact updates, or informative probing. Moreover, Eq. (3) alone does not imply that no prediction errors occur: the no-feedback condition in Observation 1 must be enforced separately. Context-independent errors can change parameters without identifying the context.

The bridge to contextual adaptation is therefore specific. Previously acquired knowledge can improve initialization; pre-contact cues or predictive deployment history can select which knowledge to use. In the absence of both useful prior structure and informative evidence, arbitrary novel dynamics cannot be identified before interaction.

2.3 Contextual prediction and composition

The desired pre-contact belief and predictive distribution are

$$\pi_{e,\tau_e^-}(c) = p(c_e = c \mid y_e, D_{<e}), \quad (5)$$

$$p(z' \mid z, a, y_e, D_{<e}) = \sum_c \pi_{e,\tau_e^-}(c) p(z' \mid z, a, c, D_{<e}). \quad (6)$$

The mixture includes a predictive hypothesis for conditions not yet represented in memory. After contact, new transitions update both context belief and, where warranted, the selected dynamics memories. These are distinct operations: *context inference* changes belief over existing hypotheses; *parameter learning* changes a hypothesis’s predictive distribution; and *memory creation* adds a hypothesis for unexplained dynamics.

For F physical factors, write $c = (c^1, \dots, c^F)$ and model

$$\mu_c(z, a) = \sum_{j=1}^F \mu_{c^j}^j(z, a) + \iota_c(z, a). \quad (7)$$

The interaction term ι_c captures departures from additive factor effects. The shared prior in Section 3.2 encodes this decomposition across complete-context memories. Prediction at an unseen combination transfers its additive component while retaining uncertainty about its unobserved interaction. We supply factor roles and cue regions as architectural structure, but no factor-value labels. Independent variation, adequate action excitation, and connected coverage of factor combinations constrain factor identification; unrestricted unsupervised disentanglement is not assumed.

3 Cira: contextual inference over residual memories

CIRA separates three operations: identifying the active physical context, learning its dynamics correction, and transferring knowledge to contexts that share physical factors. The encoder and world-model predictor remain fixed throughout. Context inference selects what the robot already knows; predictable weighted updates determine what each memory learns next.

3.1 Compact probabilistic residual memories

Let $U \in \mathbb{R}^{d \times k}$ have orthonormal columns, with $k \ll d$, and let $\phi(z, a) \in \mathbb{R}^p$ be a fixed feature map. For compact notation, this section suppresses episode indices and orders transitions by deployment time. The projected residual is $v_t = U^\top r_t$, where $r_t = z_{t+1} - f(z_t, a_t)$. Memory c predicts

$$v_{t,i} \mid c, w_{c,i} \sim \mathcal{N}\left(w_{c,i}^\top \phi_t, \sigma_i^2(z_t, a_t)\right), \quad i = 1, \dots, k, \quad (8)$$

with $\phi_t = \phi(z_t, a_t)$. The rows $w_{c,i}^\top$ form $W_c \in \mathbb{R}^{k \times p}$; its estimated latent correction is $m_c(z, a) = U \widehat{W}_c \phi(z, a)$. Each row retains a Gaussian regression state. Independent rows and diagonal, state-dependent observation noise permit inexpensive updates while representing increased variability at contact. Decorrelation of training residuals motivates the diagonal approximation but does not establish independence at deployment.

The correction basis and feature map are fixed before the evaluation stream. Residual energy outside their span measures approximation error; a small number of changing physical parameters need not induce a low-rank latent discrepancy. The full predictive distribution also includes this uncorrected error. All safety scores are evaluated in the full latent space. The analysis in Section 4 states explicitly when exact linear realizability is required.

3.2 A shared prior makes memories compositional

For context $c = (c^1, \dots, c^F)$, the prior couples memories that share factor values:

$$\text{Cov}(w_{c,i}, w_{c',i}) = k_{\mathcal{C}}(c, c') \Sigma_0, \quad (9)$$

$$k_{\mathcal{C}}(c, c') = \kappa_0^2 + \sum_{j=1}^F \kappa_j^2 \mathbf{1}\{c^j = c'^j\} + \kappa_{\times}^2 \mathbf{1}\{c = c'\}. \quad (10)$$

Here $\Sigma_0 \succ 0$ is an input-weight covariance. This separable multi-output construction [2] contains a common correction, factor-specific components, and a combination-specific interaction. It gives one coherent prior over complete context memories, including tuples never observed together. The additive factor model corresponds to $\kappa_{\times} = 0$; a positive interaction variance preserves uncertainty about unseen combinations rather than treating an additive estimate as exact.

For known combinations Ω , joint Gaussian conditioning determines a new context's prior centre and covariance. With exactly known weights, its covariance coefficient is

$$s_c = k_{\mathcal{C}}(c, c) - k_{\mathcal{C}}(c, \Omega) K_{\Omega}^{\dagger} k_{\mathcal{C}}(\Omega, c), \quad (11)$$

where K_{Ω} is the context Gram matrix and $\dagger$ denotes the Moore–Penrose inverse. Uncertain memories require the joint Gaussian conditioning formula with their observation uncertainty. Replacing this joint state by separately fitted Gaussian memory summaries is a modular approximation: correlated summaries cannot be reused as independent observations without risking overconfidence. Proposition 1 concerns the exact conditional model.

Factor roles and cue regions are specified, such as a surface region and an object region; factor-value identities are learned from dynamics without labels. Independent variation, adequate excitation, and shared state–action support make these identities distinguishable. For two additive factors, the graph of observed combinations determines whether a new pair is identifiable. The kernel hyperparameters are fixed for the analyzed evaluation stream after calibration. Their uncertainty and misspecification are tested through sensitivity and coverage measurements. A reference context is imposed as a noise-free zero correction only when that correction is known; otherwise its uncertainty is retained.

3.3 Inference before and after each transition

The hypothesis library contains encountered contexts, unseen tuples of known factors, and novel-value hypotheses. The latter retain positive support when existing factor values cannot explain the dynamics. Joint novelty must remain possible when more than one factor changes. Memory allocation and hypothesis pruning are inference approximations whose missed-context rates are measured.

Let $q_t(c)$ denote the context probability immediately before observing r_t , and $\pi_t^+(c)$ the probability afterward. At an episode boundary,

$$q_{\tau_e}(c) = \pi_{e, \tau_e^-}(c) \propto p(c \mid D_{<e}) \prod_{j=1}^F p_{\eta}(y_e^j \mid c^j). \quad (12)$$

The prior can encode frequency and persistence. During a fixed-context episode, $q_{t+1} = \pi_t^+$. A within-episode switching extension instead propagates π_t^+ through a transition model. The cue likelihood enters once per episode, before the first residual. After executing the action,

$$\pi_t^+(c) \propto q_t(c) p(v_t \mid z_t, a_t, c, \Theta_t), \quad (13)$$

using memory predictions computed before learning from that transition. This posterior supports recognition, context switching, and novelty detection. Sustained support for a new-value hypothesis allocates a memory for future observations. An unseen combination of known values already has a prior prediction and can be selected before its first contact.

Cue models are updated from a parallel context inference pass that omits the episode’s cue likelihood and uses dynamics evidence instead. Ambiguous assignments remain soft. This prevents direct training on the cue’s own pseudo-labels, although policy-dependent data and errors in stored models can still propagate. Cue reversals and appearance-only changes test whether dynamics can correct these associations.

3.4 Learning with probabilities fixed before the error

The inference posterior π_t^+ answers which context best explains the observed transition. Memory learning instead uses q_t , the probability fixed before that transition. For output row i , define

$$\omega_{c,i,t} = q_t(c) / \sigma_i^2(z_t, a_t), \quad (14)$$

$$V_{c,i,t} = P_{c,i} + \sum_{s < t} \omega_{c,i,s} \phi_s \phi_s^\top, \quad (15)$$

$$\hat{w}_{c,i,t} = V_{c,i,t}^{-1} \left(P_{c,i} \nu_{c,i,t} + \sum_{s < t} \omega_{c,i,s} \phi_s v_{s,i} \right). \quad (16)$$

The prior centre $\nu_{c,i,t}$ transfers information through Eq. (9). The positive-definite precision $P_{c,i}$ is fixed when the memory is instantiated, before its first update; a positive variance floor handles degenerate conditional priors. The centre may subsequently change using past information. The regression covariance $V_{c,i,t}^{-1}$ is distinct from the context posterior and from a guaranteed frequentist confidence region.

Only observations collected after instantiation enter these sums. Allocation-triggering residuals are not retrospectively selected into the new memory’s regression statistics. Replaying a discovery window with assignments chosen from its own outcomes defines a separate ablation. Likewise, changing the precision requires a new analysis epoch with fresh sufficient statistics and a new failure-probability allocation.

This timing matters because a weight computed from the current error can select a truncated noise distribution. Predictable weights preserve the zero-mean noise condition used by Theorem 1. They do not eliminate contamination: after a context switch, probability may still favor the preceding condition. The theorem quantifies that cost in terms of accumulated incorrect assignment weight and links it to the context filter’s log loss.

3.5 Planning and decision-aware probing

For candidate committed action sequences $\mathcal{A}$, the planner evaluates context-specific rollouts and minimizes their posterior-weighted cost,

$$\hat{J}_e(a) = \sum_c \pi_{e,\tau_e^-}(c) \hat{J}_c(a). \quad (17)$$

The analyzed planner rolls out each context’s posterior-mean correction m_c , holding the context fixed over the horizon. Corollary 2 bounds this context-weighted objective over a fixed candidate set. A separate variant samples uncertain adapter parameters once per rollout and can optimize a risk-sensitive objective. For nonlinear dynamics and costs, that parameter-averaged objective

requires its own approximation or uncertainty bound. No error observed during a committed strike can improve that strike.

A separate probe-enabled protocol permits an informative action before the consequential strike. With belief b over context and memory parameters, let $V(b) = \min_{a \in \mathcal{A}} \mathbb{E}_b[J_\theta(a)]$. The informational value of probe p is

$$\text{VOI}(p) = V(b) - \mathbb{E}_{o|p,b}[V(b_o)], \quad (18)$$

where b_o is the Bayesian belief after its outcome. Among admissible probes, the robot selects the largest positive $\text{VOI}(p) - \kappa(p)$, where κ accounts for time and task cost. Proposition 4 identifies a useful boundary: reducing context uncertainty has zero decision value when all plausible models share an optimal action.

The computational approximation reuses the current candidate costs and updates only their context weights over sampled probe outcomes. It neglects state displacement, changes in feasible strikes, and within-memory learning. When uncertainty is dominated by a single memory, a local alternative targets the variance of the cost difference between the two best actions. For one row with covariance Σ and cost-difference gradient g , its reduction is

$$\Delta_p = \frac{(g^\top \Sigma \phi_p)^2}{\phi_p^\top \Sigma \phi_p + \sigma_i^2(z_p, a_p)}. \quad (19)$$

This is a linearized decision-variance criterion, not exact value of information. Both approximations are evaluated against mutual-information probing and no probing. Probe-enabled performance is reported separately from strict first-contact retrieval.

3.6 Safety margins stored with each memory

Let $h(z) \geq 0$ define the latent safety condition, and let $\mathcal{C}_t^\delta$ be a nonempty set carrying at least $1 - \delta$ posterior mass. For every plausible memory, the safety filter checks

$$h(\bar{z}'_{c,t}) - L_h b_{c,t} \geq (1 - \alpha)h(z_t), \quad \bar{z}'_{c,t} = f(z_t, a_t) + m_{c,t}(z_t, a_t), \quad (20)$$

where L_h is a Lipschitz bound, $0 < \alpha \leq 1$, and $b_{c,t}$ is a remembered error margin. A linearized optimization can generate candidate actions, but its output requires exact constraint verification or a bounded linearization remainder to invoke Proposition 5. A stopping or retraction action counts as a safe fallback only if its true safety condition is established.

After observing z_{t+1} , score each plausible memory against its own pre-update prediction:

$$s_{c,t} = \|z_{t+1} - \bar{z}'_{c,t}\|_2. \quad (21)$$

If any plausible memory covers the transition, assign the step to the most probable one satisfying $s_{c,t} \leq b_{c,t}$. Otherwise assign it to the most probable plausible memory and record a miss. Update only this memory's raw margin,

$$b_c \leftarrow b_c + \gamma(\mathbf{1}\{s_{c,t} > b_{c,t}\} - \delta_s). \quad (22)$$

This adaptive calibration recursion [3] retains one margin per memory. The raw state is not clipped; the filter may conservatively replace it by $\max\{b_{c,t}, 0\}$. Predictive scores remain defined using the raw state for the calibration argument.

A familiar return recalls its previous margin. A wrong cue may omit the true context, but the full-state score still records the resulting mismatch. Under bounded scores and exact enforcement or a verified fallback, the calibration recursion bounds the cumulative number of one-step latent barrier failures for arbitrary cue sequences. This is a long-run accounting guarantee rather than protection against every first-contact failure. A physical guarantee additionally requires correspondence between h and the physical constraint.

3.7 Separating recall from parameter learning

Let $\pi^{\text{pre}}, \Theta^{\text{pre}}$ be the adaptive state, including shared-prior centres, immediately before assimilating the current episode’s cue, and $\pi_s^{\text{post}}, \Theta_s^{\text{post}}$ the state after s post-contact observations. Thus $s = 0$ on the updated side includes cue-based inference but no new transition. On a fixed held-out transition set $\mathcal{Q}_e$ for the current condition, evaluate four losses:

$$L_{ij}(s) = L_{\mathcal{Q}_e}(\pi^{(i)}, \Theta^{(j)}), \quad i, j \in \{0, 1\}, \quad (23)$$

where index 0 means the pre-cue snapshot and index 1 means the step- s post-cue snapshot. Posterior weights are held fixed when parameters are exchanged; inference is not rerun. The total improvement $G(s) = L_{00} - L_{11}$ decomposes as

$$R(s) = \frac{1}{2}[(L_{00} - L_{10}) + (L_{01} - L_{11})], \quad (24)$$

$$P(s) = \frac{1}{2}[(L_{00} - L_{01}) + (L_{10} - L_{11})], \quad (25)$$

$$G(s) = R(s) + P(s). \quad (26)$$

These two-player Shapley contributions average the effect of each change across the two possible orders. On return episodes with an established library, R measures the predictive contribution of recall (*apparent learning*) and P that of parameter updates (*proper learning*). A method without memory selection has $R = 0$ by construction. Either contribution can be negative, so both are reported with their signs.

Stable memory identities make the four counterfactuals comparable. When a new slot is created, its initial counterfactual uses a data-independent creation prior and zero initial selection mass; its fitted initialization cannot be inserted retrospectively into Θ^{pre} . In those windows, R is labeled *selection/creation*, since selecting an untrained slot is not recall of acquired knowledge. Creation events are reported separately, and the main recall analysis uses returns with fixed library membership. Held-out replay attributes predictive improvement conditional on the snapshots, not the causal contribution to closed-loop task success. A separate intervention that freezes all memory parameters during later returns directly tests whether retrieval alone retains the benefit.

4 Analysis: from contextual memory to first-contact control

The analysis follows the information available to the robot. First, the shared prior determines which unseen combinations can be inferred from previous experience. Second, a uniform memory-error bound quantifies what can be learned when context assignments remain uncertain. These two results supply the memory-accuracy term in the first-contact prediction bound. A simulation lemma then translates prediction accuracy into a decision gap. Recognition, probing, and safety results characterize how the same context belief is used after information becomes available.

4.1 Assumptions and scope

Let $x = (z, a) \in \mathcal{Z}_A$ denote a state-action input in the domain of analysis. Write c_t for the true context at transition t , suppressing episode indices. The memory-confidence result assumes stationary linear realizability,

$$\mu_c(x) = UW_c\phi(x), \quad v_t = W_{c_t}\phi_t + \xi_t, \quad (27)$$

with fixed U , ϕ , and W_c . Conditional on the history immediately before v_t , each $\xi_{t,i}$ is zero-mean and $\sigma_{i,t}$ -sub-Gaussian. This history includes the chosen state and action, available cues, and the

true current context for analysis; the robot does not observe that context. Contexts and actions may depend on earlier observations but not on the current noise. The noise bound $\sigma_{i,t} = \sigma_i(z_t, a_t) > 0$ and update weights are measurable before that noise is observed.

The Gaussian shared-prior model is an additional assumption for Proposition 1, not a consequence of linear realizability. The frequentist confidence result instead retains explicit prior mismatch. Define the finite separation

$$\Delta_{\max} = \sup_{c, c', x \in \mathcal{Z}_A} \|\mu_c(x) - \mu_{c'}(x)\|_2. \quad (28)$$

Realizability, stationary weights, noise domination, and bounded input domains delimit the confidence analysis. Misspecified features, underestimated noise, or unmodeled drift require additional error terms. The cost analysis uses the noise-free conditional mean map $F_c(x) = f(x) + \mu_c(x)$. The cost and safety results state their own regularity and feasibility assumptions.

4.2 When unseen combinations are identifiable

The context kernel separates two questions: whether the observed combinations identify the additive contribution of a new combination, and whether that combination has an interaction that no other context reveals. The following result makes this distinction exact in the limit in which the relevant stored predictions are known without error.

Proposition 1 (Identifiability of an unseen combination). *Fix one output coordinate and one feature vector ϕ , and let $g(c) = \phi^\top w_c$ and $\chi = \phi^\top \Sigma_0 \phi > 0$. Assume $\kappa_0^2 > 0$ and $\kappa_j^2 > 0$ for every factor. Under the context prior,*

$$g(c) = \psi(c)^\top b + \xi_c,$$

where $\psi(c)$ contains an intercept and the one-hot indicators of each factor value, b has a positive definite diagonal Gaussian covariance, and the $\xi_c \sim \mathcal{N}(0, \kappa_\chi^2 \chi)$ are independent of b and one another. Let Ω be a set of contexts with exactly observed $g(\Omega)$, including the training context, and let $q \notin \Omega$ use factor values represented in Ω . With $K_\Omega = [k_\mathcal{C}(c, c')]_{c, c' \in \Omega}$,

$$\begin{aligned} \text{Var}(g(q) \mid g(\Omega)) &= \chi s_q, \\ s_q &= k_\mathcal{C}(q, q) - k_\mathcal{C}(q, \Omega) K_\Omega^\dagger k_\mathcal{C}(\Omega, q), \end{aligned} \quad (29)$$

where $\dagger$ denotes the Moore–Penrose inverse.

(i) *If $\kappa_\chi^2 = 0$, the conditional variance is zero if and only if*

$$\psi(q) \in \text{span}\{\psi(c) : c \in \Omega\}.$$

For two factors, this holds if and only if the two factor values of q belong to the same connected component of the bipartite graph whose observed combinations are edges.

(ii) *If $\kappa_\chi^2 > 0$, then*

$$\begin{aligned} \text{Var}(g(q) \mid g(\Omega)) &= \kappa_\chi^2 \chi + \text{Var}(\psi(q)^\top b \mid g(\Omega)) \\ &\geq \kappa_\chi^2 \chi. \end{aligned}$$

Connectivity is the relevant requirement for two-factor transfer. Observing each factor value somewhere is insufficient: observations in disconnected components do not determine their relative additive contributions. A connected chain of combinations supplies those missing relations. More

observations can resolve uncertainty about the additive terms, but cannot reveal an independent interaction at a combination that has never occurred.

Equation (29) is a conditional variance, not a realized prediction error. Under the specified Gaussian model it equals the conditional mean-squared error of the conditional-mean predictor at the chosen input. A particular error may be larger or smaller. Exact recovery additionally requires an additive model and exact stored observations; finite memory uncertainty leaves additional uncertainty. The pseudoinverse is necessary because the purely additive kernel can be singular even when all its additive variance parameters are positive. If $\chi = 0$, the scalar prediction is deterministic under this prior and the span criterion is unnecessary. Fitted kernel parameters and approximate conditioning on uncertain memories provide operational estimates of this variance; they do not, by themselves, certify calibration of the fitted prior.

4.3 Memory accuracy with uncertain context assignments

Predictable assignment weights preserve the conditional zero-mean property of observation noise. They do not ensure that a sample comes from the memory being updated. The next result separates the resulting cross-context contamination from estimation noise and prior mismatch. It applies to an instantiated memory representing a stationary context; an unresolved new-value hypothesis that aggregates several contexts need not satisfy this condition.

For memory c , choose a positive definite row precision $P_{c,i}$ at creation and keep it fixed. It may depend on the information available at creation, but the sufficient statistics below contain only subsequent observations. In particular, the result does not cover selecting and replaying earlier residuals using their observed values. Write

$$\omega_{c,i,s} = \frac{q_s(c)}{\sigma_{i,s}^2}, \quad V_{c,i,t} = P_{c,i} + \sum_{s < t} \omega_{c,i,s} \phi_s \phi_s^\top, \quad (30)$$

$$\hat{w}_{c,i,t} = V_{c,i,t}^{-1} \left(P_{c,i} \nu_{c,i,t} + \sum_{s < t} \omega_{c,i,s} \phi_s v_{s,i} \right), \quad (31)$$

$$u_{c,i,t}(\phi) = \sqrt{\phi^\top V_{c,i,t}^{-1} \phi}.$$

Weights are zero before creation. The predictive probability $q_s(c)$ includes the available episode cue but excludes the current residual v_s . The prior centre $\nu_{c,i,t}$ may be recomputed from the observation history; this flexibility does not extend to changing $P_{c,i}$ while retaining these sufficient statistics.

Theorem 1 (Uniform memory error under context uncertainty). *Under the stationary realizability and conditional sub-Gaussian noise assumptions, fix a memory c and a failure probability $\delta_c \in (0, 1)$. Conditional on the information at its creation, with probability at least $1 - \delta_c$, simultaneously for every subsequent t , every $\phi \in \mathbb{R}^p$, and every output coordinate $i \in \{1, \dots, k\}$,*

$$\left| \phi^\top (\hat{w}_{c,i,t} - w_{c,i}) \right| \leq \beta_{c,i,t} u_{c,i,t}(\phi), \quad (32)$$

where

$$\beta_{c,i,t} = \sqrt{2 \log \frac{k}{\delta_c} + \log \frac{\det V_{c,i,t}}{\det P_{c,i}}} + \|\nu_{c,i,t} - w_{c,i}\|_{P_{c,i}} + B_{c,i,t}, \quad (33)$$

$$B_{c,i,t}^2 = \sum_{s < t} \omega_{c,i,s} \left[\phi_s^\top (w_{c_s,i} - w_{c,i}) \right]^2. \quad (34)$$

Here c_s is the true context. Assigning failure probabilities with $\sum_c \delta_c \leq \delta$ makes these inequalities simultaneous over all instantiated memories with probability at least $1 - \delta$.

If all observations lie in the domain defining $\Delta_{\max}$, the contamination is bounded by

$$B_{c,i,t}^2 \leq \Delta_{\max}^2 M_{c,i,t}, \quad M_{c,i,t} = \sum_{s < t: c_s \neq c} \omega_{c,i,s}.$$

The noise term follows the self-normalized martingale bound for predictable linear regression [1]. Predictability is the essential requirement: the feature, noise scale, and assignment weight must be fixed before the current noise is observed. Neither independence between context assignments and earlier observations nor independent output coordinates is required. The simultaneous statement over inputs permits evaluation on the candidate trajectories considered by the planner. Fixed feature maps, fixed correction directions, and fixed per-memory precisions are part of the scope of this result.

The contamination term connects memory accuracy to context prediction. If every noise scale is at least $\sigma_{\min} > 0$, then, for each output coordinate,

$$\sum_c M_{c,i,T} \leq \frac{1}{\sigma_{\min}^2} \sum_{t < T} [1 - q_t(c_t)] \leq \frac{1}{\sigma_{\min}^2} \sum_{t < T} \ell_t^{\text{dyn}}, \quad \ell_t^{\text{dyn}} = -\log q_t(c_t). \quad (35)$$

The sum on the left includes only instantiated memories. If the true context has no represented slot, set its probability to zero; the log-loss bound is then infinite and supplies no finite claim for that step. Unresolved new-value slots are not stationary memories covered by the theorem. Once contexts are represented, accurate predictions reduce both the error of selecting a memory and the contamination of memories updated with uncertain assignments.

Theorem 1 is an error bound, not a claim that an uninflated Gaussian posterior interval has nominal coverage. Both the prior-mismatch norm and the contamination term involve unknown quantities. A computable certified radius requires valid upper bounds on these terms and on the noise scale; substituting fitted estimates does not preserve the coverage guarantee automatically. Known parameter ranges in state-based simulations can supply conservative envelopes. Empirical coverage evaluates fitted intervals when certified envelopes are unavailable. Negative transfer appears explicitly as a large prior-mismatch term. Additional data reduce $u_{c,i,t}(\phi)$ only in directions sufficiently explored by the weighted features, and persistent contamination need not vanish with sample size.

For use in the first-contact bound, on the simultaneous event a valid vector error envelope is

$$\epsilon_t = \sup_{c, (z,a) \in \mathcal{Z}_A} \left(\sum_{i=1}^k \beta_{c,i,t}^2 u_{c,i,t}(\phi(z,a))^2 \right)^{1/2},$$

with the supremum restricted to the memories under consideration. Orthonormality of U transfers this projected bound to the latent correction. A finite, useful envelope requires bounded prior mismatch, controlled contamination, and adequate feature coverage.

Why filtered weights can introduce bias. Consider scalar observations $v \sim \mathcal{N}(0, \sigma^2)$ and two fixed scoring models with means 0 and $\Delta > 0$, equal variances, and equal predictive probabilities. Hard assignment by the filtered probability sends an observation to the first model exactly when $v < \Delta/2$. A separate sample-mean estimator trained on those selected observations converges to

$$\mathbb{E}[v \mid v < \Delta/2] = -\sigma \frac{\varphi_{\mathcal{N}}(\Delta/(2\sigma))}{\Phi_{\mathcal{N}}(\Delta/(2\sigma))} < 0,$$

where $\varphi_{\mathcal{N}}$ and $\Phi_{\mathcal{N}}$ are the standard normal density and distribution function. At $\Delta = 2\sigma$, the bias is approximately -0.29σ . An interval that shrinks around this selected-sample mean eventually

excludes zero. This counterexample keeps the scoring models fixed; it does not assert the same limiting value when those scoring models are learned online. Predictable weights remove this dependence on the current noise, while the contamination term accounts for samples generated by other contexts.

4.4 First-contact error is controlled by memory accuracy and context log loss

For a common query $x_e = (z_{e,\tau_e}, a_{e,\tau_e})$, define the mean prediction bias

$$b_e^{\text{pred}} = \left\| \sum_c \pi_e(c) m_{c,e}(x_e) - \mu_{c_e}(x_e) \right\|_2, \quad \pi_e(c) = \pi_{e,\tau_e^-}(c).$$

This quantity compares conditional means; it excludes irreducible observation noise. The error of an observed transition additionally includes that noise.

Proposition 2 (First-contact separation). *Consider a closed library of K contexts containing each current context, and suppose $\|m_{c,e}(x) - \mu_c(x)\|_2 \leq \epsilon_e$ for all supported contexts and queries under consideration. Then*

$$b_e^{\text{pred}} \leq \epsilon_e + (1 - \pi_e(c_e)) \Delta_{\max}, \quad (36)$$

$$\sum_{e=1}^N b_e^{\text{pred}} \leq \sum_{e=1}^N \epsilon_e + \Delta_{\max} \sum_{e=1}^N -\log \pi_e(c_e). \quad (37)$$

By contrast, under Observation 1, an episode-reset predictor initialized at f has bias $\|\mu_{c_e}(x_e)\|_2$. If this norm is at least $\Delta_{\min} > 0$ on N_{shift} evaluated episodes, its cumulative bias is at least $N_{\text{shift}} \Delta_{\min}$.

The upper bound for contextual prediction is smaller than the reset lower bound whenever

$$\sum_e \epsilon_e + \Delta_{\max} \sum_e -\log \pi_e(c_e) < N_{\text{shift}} \Delta_{\min}. \quad (38)$$

This is a sufficient condition for separation on the specified queries, not a guarantee for every deployment. Theorem 1 supplies ϵ_e on its simultaneous confidence event when the inputs and contexts satisfy its assumptions. The context log loss has two roles: episode-level loss controls anticipatory selection, while transition-level loss controls contamination in Eq. (35).

The full method also retains a novel-context branch. If $p_\star$ is its aggregate mass, $m_\star$ its conditional mean, and the true context is already known, the corresponding one-query bound is

$$b_e^{\text{pred}} \leq (1 - p_\star) \epsilon_e + (1 - p_\star - \pi_e(c_e)) \Delta_{\max} + p_\star \|m_\star(x_e) - \mu_{c_e}(x_e)\|_2. \quad (39)$$

Without a bound for the last term, a closed-library guarantee does not apply to the entire mixture. Truly novel conditions require an acquisition phase or a valid transferred prior; no fixed one-episode acquisition cost is assumed.

Corollary 1 (Finite-alphabet entropy reference). *Suppose the context alphabet of size K and cue alphabet $\mathcal{Y}$ are fixed, and each episode’s context identity is revealed after its prediction. For each cue value, use a categorical predictor with symmetric Dirichlet-1/2 counts:*

$$\pi_e^{\text{KT}}(c | y) = \frac{n_{<e}(c, y) + 1/2}{n_{<e}(y) + K/2}.$$

For any context–cue sequence, its cumulative log loss satisfies

$$\sum_{e=1}^N -\log \pi_e^{\text{KT}}(c_e \mid y_e) \leq N\hat{H}(C \mid Y) + O(K|\mathcal{Y}|\log(N+1)). \quad (40)$$

Substitution into Eq. (37) gives the corresponding first-contact bias bound for this reference predictor.

This is the categorical coding bound of Krichevsky and Trofimov [4], stated separately because its feedback assumptions differ from unlabeled context discovery. CIRA receives no context labels, and its inferred assignments do not automatically inherit Eq. (40). Empirical conditional entropy therefore diagnoses available cue information, while measured context log loss tests the inference procedure. Neither quantity alone determines prediction error without memory accuracy and residual separation.

4.5 Recognition depends on initial evidence and dynamics separation

For a known true context c and one fixed alternative c' , let

$$\ell_0 = \log \frac{p(c \mid D_{<e})}{p(c' \mid D_{<e})} + \log \frac{p_\eta(y_e \mid c)}{p_\eta(y_e \mid c')}$$

be the pre-contact log odds. A diagnostic observation contributes a log-likelihood ratio in favor of c .

Proposition 3 (Two-context recognition time). *Assume fixed likelihoods and independent, identically distributed diagnostic observations under c , with integrable log-likelihood increments of mean $D > 0$. Let T be the first time the accumulated log odds reach a threshold $\ell_\star$, and assume $\mathbb{E}[T] < \infty$ and integrable threshold overshoot O_T . If $\ell_0 \geq \ell_\star$, then $T = 0$. Otherwise,*

$$\mathbb{E}[T] = \frac{\ell_\star - \ell_0 + \mathbb{E}[O_T]}{D} \geq \frac{\ell_\star - \ell_0}{D}. \quad (41)$$

For correctly specified likelihoods, D is their Kullback–Leibler divergence under the diagnostic observation distribution.

The result isolates how a cue can replace post-contact evidence. Larger initial log odds reduce the threshold deficit, while better separated dynamics increase the evidence rate. Repeated exposure can improve both cue associations and memory likelihoods. Occurrence counts alone need not produce savings when competing contexts recur equally often. Adaptive actions, changing memories, multiple alternatives, and persistence-based recognition criteria require additional analysis; the two-context experiment controls these factors to test Eq. (41).

4.6 From model accuracy to the committed decision

Prediction error matters through the action selected before contact. We connect the two for a fixed context over a noise-free horizon of H transitions. A candidate $a = (a_0, \dots, a_{H-1})$ is a committed action sequence, and its cost is

$$J_e(a) = \sum_{t=0}^{H-1} \ell(z_t^a, a_t) + g(z_H^a), \quad (42)$$

where $g = 0$ permits a problem without terminal cost. Assume the true map F_{c_e} is L -Lipschitz in state, uniformly over candidate actions, and $\ell(\cdot, a_t)$ and g are L_ℓ -Lipschitz. These conditions hold on a domain containing both true and predicted trajectories. Set $C_H = \sum_{j=0}^{H-1} L^j$.

Lemma 1 (Simulation bound). *Fix a candidate sequence a and let z_t and $\hat{z}_t$ be the noise-free trajectories of F_{c_e} and a model $\hat{F}$, respectively, from the same initial state. Under the preceding Lipschitz assumptions,*

$$|J_e(a) - \hat{J}(a)| \leq L_\ell C_H \sum_{t=0}^{H-1} \|F_{c_e}(\hat{z}_t, a_t) - \hat{F}(\hat{z}_t, a_t)\|. \quad (43)$$

The discrepancy is evaluated along the model’s trajectory. That trajectory is available during planning; bounding its discrepancy additionally requires a valid error bound. The statement concerns fixed action sequences. Feedback policies require corresponding regularity assumptions on their closed-loop dynamics and costs.

Corollary 2 (First-contact decision gap). *Let $\mathcal{A}_e$ be a finite, common candidate set and write $\pi_e(c) = \pi_{e, \tau_e^-}(c)$ for a belief over a closed library containing the true context c_e . Let $\hat{J}_{e,c}(a)$ roll out $f + m_c$, and set*

$$\hat{J}_e(a) = \sum_c \pi_e(c) \hat{J}_{e,c}(a), \quad \hat{J}_e(\hat{a}_e) \leq \min_{a \in \mathcal{A}_e} \hat{J}_e(a) + \eta_{\text{opt},e}. \quad (44)$$

Suppose $\|m_c(x) - \mu_c(x)\| \leq \bar{\epsilon}_e$ for every supported context and every input along its candidate rollouts. Suppose also that $\|\mu_c(x) - \mu_{c_e}(x)\| \leq \Delta_{\max}$ there. Then

$$J_e(\hat{a}_e) - \min_{a \in \mathcal{A}_e} J_e(a) \leq 2L_\ell C_H H [\bar{\epsilon}_e + (1 - \pi_e(c_e))\Delta_{\max}] + \eta_{\text{opt},e}. \quad (45)$$

With common constants across episodes,

$$\sum_{e=1}^N \left[J_e(\hat{a}_e) - \min_{a \in \mathcal{A}_e} J_e(a) \right] \leq 2L_\ell C_H H \left[\sum_{e=1}^N \bar{\epsilon}_e + \Delta_{\max} \sum_{e=1}^N -\log \pi_e(c_e) \right] + \sum_{e=1}^N \eta_{\text{opt},e}. \quad (46)$$

The comparison is with the best candidate, and the uniform error requirement includes candidates the planner does not select. The bound separates memory accuracy, context selection, and optimization error. It can be loose over long horizons: when $L = 1$, $C_H = H$. Quantitative use therefore requires validated dynamics and cost regularity; the statement does not extend automatically to discontinuous contact costs or stochastic rollouts.

4.7 When probing changes the decision

A probe is useful when its observations improve the ensuing decision enough to cover its cost. To isolate this information value, fix the state, a finite action set $\mathcal{A}$, and integrable costs $J_\theta(a)$, where $\theta = (c, W)$ includes context and memory parameters. For belief b and the Bayesian posterior b_o after outcome o of probe p , define

$$V(b) = \min_{a \in \mathcal{A}} \mathbb{E}_b[J_\theta(a)], \quad \text{VOI}(p) = V(b) - \mathbb{E}_o[V(b_o)]. \quad (47)$$

Proposition 4 (Information value and action agreement). *Under these fixed-state, fixed-action-set conditions, every probe has $\text{VOI}(p) \geq 0$. If one action $a^\dagger$ minimizes J_θ for b -almost every θ , then $\text{VOI}(p) = 0$ for every probe. Nevertheless, $I(\theta; o | p)$ can be positive: for a discrete belief this occurs whenever two states of positive probability have different probe-outcome distributions.*

Action disagreement is necessary, but not sufficient, for positive decision value. With nonnegative probe cost $\kappa(p)$, a rule requiring $\text{VOI}(p) > \kappa(p)$ never selects a probe under common action agreement. Information gain alone can reward observations that leave the optimal decision unchanged.

The context-reweighting score used in planning approximates this criterion by holding memory parameters and the initial state fixed. A physical nudge can also change the state, refine a memory, or change which strikes pass the safety filter. Those effects alter the decision problem and are outside Proposition 4. Likewise, local reduction in the variance of a linearized cost difference measures decision-relevant uncertainty, but is not an exact value-of-information calculation.

4.8 Reusing calibrated latent safety margins

Calibration uses the full latent prediction error, including noise, context mistakes, and residuals outside the correction subspace. Let $\mathcal{C}_t^\delta$ be the nonempty plausible set and $\bar{z}'_{c,t}$ the prediction computed before observing z_{t+1} . Its score is $s_{c,t} = \|z_{t+1} - \bar{z}'_{c,t}\|$. With an L_h -Lipschitz barrier function h and $0 < \alpha \leq 1$, the filter checks

$$h(\bar{z}'_{c,t}) - L_h b_{c,t} \geq (1 - \alpha)h(z_t), \quad c \in \mathcal{C}_t^\delta. \quad (48)$$

Each step is attributed to exactly one memory: a covered memory satisfying $s_{c,t} \leq b_{c,t}$ if any exists, and otherwise the most probable plausible memory. Ties are immaterial. Only its margin is updated, without projection, by

$$b_{c,t+1} = b_{c,t} + \gamma(\mathbf{1}\{s_{c,t} > b_{c,t}\} - \delta_s). \quad (49)$$

Proposition 5 (Long-run latent barrier violations). *Assume Eq. (48) is enforced exactly whenever feasible; otherwise a fallback satisfies $h(z_{t+1}) \geq (1 - \alpha)h(z_t)$. Suppose all calibration scores lie in $[0, \bar{S}]$, each margin is initialized in $[0, \bar{S}]$, $\gamma > 0$, and $0 < \delta_s < 1$. Let K_T count all calibration streams initialized during the first T steps, including resets. With the attribution and update above, for every sequence of contexts and cues,*

$$\sum_{t=0}^{T-1} \mathbf{1}\{h(z_{t+1}) < (1 - \alpha)h(z_t)\} \leq \delta_s T + K_T \frac{\bar{S} + 2\gamma}{\gamma}. \quad (50)$$

This pathwise calibration bound follows the deterministic adaptive conformal argument [3]; it does not require a correct context posterior or a Gaussian noise model. Its assumptions still require bounded scores and a safe fallback. An asymptotic violation rate of at most δ_s requires $K_T/T \rightarrow 0$. Repeatedly resetting margins consumes the same allowance as creating additional streams.

The counted event is failure of the one-step latent barrier inequality. It is not the number of time steps spent outside the safe set, and a per-episode rate requires one calibrated decision per episode or an episode-level score. Physical interpretation depends on how h represents the physical constraint. Finally, a quadratic program obtained by linearizing the filter must verify the nonlinear inequality, or include a valid remainder bound, to meet the exact-enforcement assumption.

5 Evaluation protocol

The evaluation tests when information available before interaction changes the first consequential prediction, whether this improvement comes from recalling physical knowledge, and how context uncertainty affects confidence and control. Composition, cue reliability, and recurrence are controlled independently. Numerical outcomes are unmeasured and denoted **XX.XX**; the protocols below specify the comparisons and endpoints.

5.1 Tasks, information, and evaluation protocol

Task families. Controlled simulation uses puck striking and pushing near an edge, with independently manipulated surface, load, and actuator conditions. Visual-world-model benchmarks provide broader model and task coverage; tasks permitting repeated correction are identified separately from strict single-contact tasks. The hardware protocol prioritizes puck striking and may include initial load lifting, evaluating the force or motion committed before load-dependent feedback can affect it. The protocol first verifies that each physical manipulation changes the chosen interaction observably: for example, ideal Coulomb sliding deceleration alone does not identify mass. Backbones, adapter features, and hyperparameters are fixed using development conditions and streams disjoint from evaluation.

The first-contact endpoint. Let τ_e mark action commitment before the first informative interaction in episode e . The primary prediction metric is the mean squared residual error per latent dimension,

$$\mathcal{E}_{\text{first}}^e = \frac{1}{d} \|r_{e,\tau_e} - \hat{\mu}_{e,\tau_e}(z_{e,\tau_e}, a_{e,\tau_e})\|_2^2, \quad r_{e,t} = z_{e,t+1} - f(z_{e,t}, a_{e,t}), \quad (51)$$

where $\hat{\mu}$ is the predicted mean residual and the prediction is recorded before observing the resulting transition. Because a contact transition can be a poor proxy for the decision that matters, we also predefine a rollout endpoint, such as puck stopping position, and measure its prediction error, task cost, and success under the committed action. Probe actions are excluded from these tests. Prediction comparisons use shared held-out state-action replay; separate closed-loop trials measure each controller’s realized outcomes. The replay set never updates memories. Latent errors are only compared within a common fixed representation. Native methods that change their encoder are also evaluated through physical endpoint error and task outcomes, with explicitly labeled predictor-only variants used for matched latent-error analyses.

Proposition 2 concerns the norm of the conditional mean error, rather than the realized squared error above. State-based simulation therefore also records $\|\hat{\mu}_e(x) - \mu_{c_e}(x)\|_2$ using the simulator’s known mean dynamics. This diagnostic separates noise from prediction bias and permits direct comparison with the theoretical bound. Decision-gap measurements use the same finite action set for the learned planner and the true-model optimizer; the noise-free, fixed-horizon analysis of Corollary 2 is evaluated separately from stochastic task performance.

Information and replication. Methods share sensory inputs, episode boundaries, backbone initialization where compatible, action budgets, and planning budgets. Simulator parameters, context labels, future schedules, and identifying metadata are unavailable except to declared oracles and evaluation. Streams contain N episodes and are replicated over n_{seed} independent schedules/seeds, paired across methods. Confidence intervals use independent streams as the sampling unit; hardware analyses account for session dependence. We report paired effects, uncertainty, and all predefined conditions rather than select favorable stream segments. Runtime and total stored state accompany accuracy comparisons.

Cue and history controls. For balanced K -context tests, a symmetric channel emits the matching cue with probability ρ and each other cue with probability $(1 - \rho)/(K - 1)$; chance is $1/K$. We include the $\rho = 0.8$ condition and report empirical $H(C | Y)$ alongside cue and posterior log loss. Balanced randomized episode contexts isolate cues from history; persistent schedules separately test informative recurrence priors. A fixed episode cue is assimilated once. Cue associations are learned from dynamics-based assignments, without physical labels. All-novel conditions and repeated known

Family	Across-episode state	Pre-contact evidence	Role
Frozen	Fixed backbone	Common observation	No adaptation
Episode adaptation	None after reset	Common observation	Reactive updates
Continual adaptation	Shared parameters; replay if declared	History and common observation	Persistence without separate memories
Changepoint or continuous context	Declared retained state	History; cues for cue-enabled variant	Alternative contextual adaptation
Dynamics-only mixture	Separate dynamics memories	History prior	Memory without cue association
Cira ablations/full	Memories and learned associations	History; cues when enabled	Isolate cues, persistence, factors, noise, assignment
Oracle retrieval	Same learned memories	True context or factor IDs	Isolate inference error

Table 1: Baseline information access. Common observations may already contain appearance information; a dynamics-only method disables the additional learned cue-to-memory association. Oracle retrieval receives neither perfect residual parameters nor extra transitions.

conditions are reported separately. A separate finite-alphabet oracle diagnostic supplies past episode labels only to the sequential predictor in Corollary 1; its entropy comparison is not treated as a guarantee for inferred labels or growing libraries.

5.2 Baselines organized by retained information

Table 1 specifies the information each family can retain. Only comparisons relevant to an experiment are run. Episode-reset gradient and residual baselines use AdaJEPA-style and Sandwich-Residuals-style updates where technically compatible [6, 7]; implementation differences and reset policies are explicit. A dynamics-only mixture follows the persistent-mixture setting [5]. A persistent cue-conditioned encoder and a continuous-context Gaussian process with a state-action kernel multiplied by an environment-descriptor kernel test whether benefits require discrete memories. The continuous descriptor combines the cue with the same recent transition window available to Cira.

5.3 Controlled experiments

5.3.1 Experiment 1: The first-contact gap

Hypothesis. Informative cues improve first-contact prediction when they retrieve previously learned dynamics, whereas a reset method receiving no current-episode update before commitment retains its initialization’s prediction.

Manipulation and methods. After a common calibration stream, we randomize balanced recurring contexts and sweep ρ . Comparisons include frozen, reset gradient and residual adaptation, continual adaptation, a dynamics-only mixture, Cira without cues, a persistent cue-conditioned encoder, full Cira, and oracle-context Cira. Identical initialization and planning support an explicit numerical equality check between frozen and reset predictions before feedback. No-cue memory methods may still benefit from a useful prior; the balanced schedule controls the contribution of predictive temporal order and unequal frequencies.

Readout and falsification. Figure 2 plots first-contact residual and committed-endpoint error against cue reliability, with an information-axis companion and success estimates. The first-contact effect at $\rho = 0.8$ is **XX.XX** relative to the strongest relevant baseline. The simulator-based mean-error diagnostic overlays the bound from Proposition 2 using the recorded pre-contact posterior and independently assessed memory error. No improvement over cue-free inference under informative cues would weaken pre-contact retrieval. A discrepancy in the reset equality check triggers an information or initialization audit, not a claim against the timing observation.

5.3.2 Experiment 2: Recurrence is the resource

Hypothesis. Persistent memories become useful as previously learned contexts recur; no comparable advantage is assumed when every condition is novel and lacks transferable structure.

Manipulation and methods. We vary realized reuse frequency from all-novel streams to frequent returns while holding total transition budget, physical range, task difficulty, and cue channel fixed. Return lag is varied separately. Cira, cue-free Cira, continual adaptation, and reset residual adaptation are compared. Cold-start streams quantify the cost of acquiring memories; an additional matched-calibration experiment isolates reuse from differences in training data per context. Repertoire size and exposure counts are reported.

Readout and falsification. A Figure 2 inset shows cumulative first-contact regret or error against reuse, with novel and returning episodes separated. Success and memory acquisition cost accompany the curves. Failure to gain on returns after adequate calibration would weaken recurrence as the mechanism. Benefits in the all-novel regime would require an explanation through shared structure or other adaptation effects, rather than being attributed to recall.

5.3.3 Experiment 3: Savings curves

Hypothesis. Repeated exposure can reduce recovery time by improving memories and cue associations; strong initial evidence can permit recognition before contact.

Manipulation and methods. Sequences $A \rightarrow B \rightarrow A \rightarrow B \rightarrow A$ vary cue strength with balanced exposure and common post-contact diagnostic actions. Reset gradient, continual gradient, dynamics-only contextual inference, and full Cira are compared. Contextual recognition requires posterior probability at least $1 - \delta_{\text{rec}}$ under a predefined persistence criterion. A shared held-out prediction-loss recovery threshold permits comparison with gradient methods that have no discrete context identity.

Readout and falsification. Figure 3 shows recognition/recovery delay against occurrence, including zero-delay cases and censored failures, and delay against initial cue log odds. Proposition 3 is assessed in a controlled two-context test with fixed memories and independent stationary diagnostic observations. We estimate the mean log-likelihood increment D , record threshold overshoot, and compare observed stopping times with the corresponding hitting-time prediction. The separate persistence-based recognition endpoint tests robustness in closed-loop trajectories, where the idealized increment assumptions need not hold. Increasing occurrence does not necessarily increase prior odds when competing counts grow together; no unconditional occurrence law is assumed. No improvement after memories or cue associations improve, or failure of initial evidence to explain delay under these controls, would weaken the savings account.

5.3.4 Experiment 4: Memory versus optimization

Hypothesis. Later returns derive more of their prediction improvement from retrieval than first exposures do.

Manipulation and methods. We snapshot (π, Θ) immediately before the current cue and again after s observations. The four replay losses $L_{ij}(s)$ evaluate old/new posterior and old/new memory parameters on one fixed held-out transition set. Their signed Shapley contributions yield recall $R(s)$ and parameter learning $P(s)$ with $G(s) = R(s) + P(s)$. Memory slots are aligned across snapshots, using declared pre-creation priors for new slots. Full Cira is compared with a post-calibration intervention that freezes all memory weights and shared prior centers, and with reset adaptation whose recall term is zero.

Readout and falsification. Figure 3 plots signed contributions for first, second, and later occurrences. Ratios are omitted when total gain is near zero or negative. These quantities attribute snapshot changes; they do not independently prove a causal training mechanism. Little return-episode gain with frozen weights, or continuing dominance of parameter changes, would weaken the claim that retrieval changes adaptation’s mechanism.

5.3.5 Experiment 5: Composition by factorial holdout

Hypothesis. The shared factor kernel supports first-contact prediction for an unobserved combination when seen combinations identify its additive effects and interaction uncertainty is represented adequately.

Manipulation and methods. We observe $A1, A2, B1, B3, C2, C3$ and hold out $A3, B2, C1$. Every factor value is therefore experienced, and the observation graph is connected. Factor-local cue regions, independent interventions, and sufficient excitation provide structure for discovering factor values without labels. A disconnected control observes $A1, A2, B1, B2, C3$ and holds out a cross-component pair such as $A3$; factor exposure and transition budgets are matched where possible. Held-out combinations are excluded from learning and tuning; each test restarts from the same pretest checkpoint.

Comparisons include independent whole-context memories, the additive shared kernel with $\kappa_{\times} = 0$, the full shared kernel with $\kappa_{\times} > 0$, the continuous-context baseline, and oracle factor selection. Every method receives the same observed combinations. Before evaluation, each shared model records its conditional variance for every held-out tuple using only the training combinations and fixed hyperparameters. A controlled interaction-strength sweep λ complements realistic nonadditive physics. A separate prior-matched simulation checks Proposition 1 in the limit of accurately observed context means.

Readout and falsification. Figure 4 shows the observation graph, first-ever-contact prediction and task outcomes for each held-out cell, and error against λ . The model’s conditional variance is a predicted uncertainty, not a deterministic bound on an individual squared error. Under the prior-matched simulator, repeated draws compare this variance with mean squared error of the true residual mean; noisy outcome error is reported separately. The connected/disconnected contrast tests additive identifiability, while nonzero $\kappa_{\times}$ tests the irreducible uncertainty of an unseen interaction. Post-contact adaptation requirements are secondary. Gains appearing only after observing the

held-out combination would not support zero-shot composition; overconfidence or negative transfer under strong interactions would qualify the shared-prior claim.

5.3.6 Experiment 6: Cue corruption and causal conflict

Hypothesis. Cues act as revisable probabilistic evidence, with informative dynamics able to override misleading appearance.

Manipulation and methods. We test independent cue shuffling, partial reliability, learned cue reversal, novel appearance with known physics, and familiar appearance with novel physics. Digital or camera-side masking hides appearance without changing the contact surface. Full Cira, cue-free inference, and hard cue/MAP assignment share memory initialization. Cue updates use dynamics-derived assignments; one fixed cue is not counted repeatedly.

Readout and falsification. A posterior timeline around conflict displays pre-contact beliefs and their revision. Metrics include first-contact error, cue override samples, posterior log loss, false memory creation, memory contamination, and task/safety cost. Wrong cues may worsen the committed first action; no universal immunity is claimed. Persistent confidence in the wrong memory despite diagnostic transitions, or systematic contamination that prevents later recall, would weaken probabilistic cue integration. Incorrectly creating new memories for appearance-only changes would reveal another distinct failure.

5.3.7 Experiment 7: Context boundary stress test

Hypothesis. State-dependent uncertainty distinguishes transient contact noise from persistent dynamics changes more effectively than a constant-noise likelihood.

Manipulation and methods. We cross true context separation with noise magnitude at impact, stick-slip, and transient acceleration. Unchanged-context noisy trials and true-switch trials are both included. Constant-noise Cira, heteroscedastic Cira, and an explicit changepoint baseline use matched development data and independently tuned detection thresholds.

Readout and falsification. Heat maps show false creation and missed-change rates across separation and noise. Detection delay, predictive coverage, and sensitivity tradeoff curves prevent a method from appearing better merely by refusing to create memories. False creations are normalized by unchanged-context exposure. A reduction in false splits obtained only by missing true changes or greatly delaying detection would not support the uncertainty-model claim. Similar performance at matched sensitivity and delay would make heteroscedasticity an implementation choice rather than a scientific result.

5.3.8 Experiment 8: Continual interference

Hypothesis. Separately retained memories preserve useful predictions for A as intervening B experience grows.

Manipulation and methods. After matched calibration on A, we vary B duration across logarithmically spaced transition budgets before returning to identical A test states. Continual gradients, periodic resets, a replay baseline where feasible, and Cira are compared. Both full retrieval and oracle selection of the stored A memory are evaluated, separating retrieval-prior decay from parameter corruption. Soft memory updates remain active unless explicitly ablated.

Readout and falsification. A small Figure 3 panel plots immediate A-return error against B exposure, with endpoint success as a companion result. Deterioration under oracle A selection would indicate interference in stored knowledge; deterioration only under inferred selection would indicate retrieval failure. Either would qualify the preservation claim. We do not infer noninterference solely from the existence of separate parameter arrays.

5.3.9 Experiment 9: Safety as contextual reuse

Hypothesis. Recalling context-specific calibration can reduce unnecessary intervention on familiar returns at comparable observed violation rates.

Manipulation and methods. Filters use a fixed/global margin, an adapted shared margin, all stored memories, Cira’s posterior-selected models with per-memory margins, or oracle context. Dynamics memories and candidate actions are otherwise matched. Evaluation includes familiar returns, ambiguity, novel physics, and misleading cues. Each memory’s score is the full latent distance from its own prediction, and only the attributed memory receives the ACI margin update. Operating points are selected on development streams, not adjusted retrospectively on test outcomes. The simulator records the score bound $\bar{S}$, the number of created slots K_T , and every fallback event needed to assess Proposition 5.

Readout and falsification. Figure 5 shows violation–intervention and violation–cost tradeoffs, action modification, fallback frequency, and margin recovery after switches. Confidence intervals accompany rare-event rates. The cumulative number of violations of the one-step latent condition is compared with $\delta_s T + K_T(\bar{S} + 2\gamma)/\gamma$ only where the bounded-score, exact-filter, and safe-fallback assumptions hold. This diagnostic includes misleading cues even when the plausible set excludes the true context; the guarantee concerns calibration of the actual attributed prediction error. Failure of an assumption is reported separately from violation of the bound. Latent-condition violations and independently observed physical violations remain separate endpoints. No reduction in conservatism at comparable supported risk would weaken the contextual-calibration claim; an average violation bound does not establish physical safety or a bound on every individual episode.

5.3.10 Experiment 10: Long-stream deployment

Hypothesis. An uninterrupted deployment develops a reusable repertoire whose benefits remain visible at individual context returns.

Manipulation and methods. The hardware protocol targets at least 1000 episodes if hardware permits, with contexts recurring on a schedule hidden from the robot. The stream includes novel conditions and controlled cue changes. Memories persist throughout; resets, maintenance, failures, and day boundaries are logged. Independent simulation streams support paired baseline comparisons; available hardware sessions determine the scope of hardware uncertainty estimates.

Readout and falsification. A complete temporal visualization aligns true conditions, inferred contexts, memory count, first-contact error, cue calibration, recall/learning contributions, success, and margins/violations. The main text summarizes return-event trajectories and occurrence-conditioned performance, not only an average; the full timeline remains inspectable. Duplicate-memory growth, declining return performance, or repeated relearning despite familiar conditions would weaken the repertoire claim. An isolated hardware stream is presented as a deployment demonstration rather than independent replication of every comparison.

5.3.11 Experiment 11: Confidence under uncertain assignments

Hypothesis. Predictable pre-error weights support the confidence guarantee in Theorem 1; posterior weights that depend on the current residual can introduce selection bias. Context contamination and prior mismatch remain separate sources of error under either update rule.

Manipulation and methods. Controlled linear simulations vary context separation relative to noise, cue reliability, switch frequency, and shared-prior misspecification. The principal comparison uses predictive weights $q_t(c)$ and filtered weights $\pi_t^+(c)$ with identical features, noise scales, and memory initializations. Hard filtered assignment and oracle context assignment provide additional controls. Noise scales are known in the guarantee check; learned and deliberately underestimated scales constitute separate stress tests. New memory slots begin the confidence analysis after their creation, without retroactively using the residual that triggered creation as a predictably assigned observation.

The diagnostic records each coordinate’s total weight $n_{c,i,T} = \sum_{t < T} \omega_{c,i,t}$, contaminated weight $M_{c,i,T} = \sum_{t < T: c_t \neq c} \omega_{c,i,t}$, and prior mismatch. Simulator labels and true weights are used only to calculate these diagnostics. Oracle confidence radii use the true contamination and prior terms, or valid predetermined upper bounds; plug-in radii based on learned separation estimates are labeled empirical intervals and evaluated separately. The scalar fixed-gating example accompanying Theorem 1 isolates residual-dependent selection bias before the more general adaptive-filter comparison.

Readout and falsification. Coverage concerns the true mean $w_{c,i}^\top \phi$, not a fresh noisy transition. Across independent streams we report both pointwise coverage and the fraction of streams with any failure on the predefined finite query/time grid. This finite-grid check can detect a failure of the simultaneous guarantee but cannot verify its claim over all inputs. Interval width, bias, contamination fraction, and coverage are stratified by contact phase and context overlap. Prediction intervals for noisy outcomes include observation noise and are reported separately. Undercoverage with valid oracle terms challenges the implementation or assumptions; undercoverage confined to plug-in intervals identifies an insufficient empirical radius. The theorem does not require every filtered-weight implementation to fail, and the experiment does not assume such a universal outcome.

5.3.12 Experiment 12: Decision-aware probing

Hypothesis. A probe can inform the context without improving the strike decision. Decision-aware selection avoids paying for such probes and gains when their outcomes change the preferred strike enough to offset probe cost.

Manipulation and methods. The probe protocol is separate from the no-probe first-contact evaluation. It compares no probing, context-information maximization, value-of-information selec-

tion, and the linearized c -optimal criterion on the same candidate probes and strike set. Every probe satisfies the same safety checks and counts against the same interaction budget. The probe cost $\kappa(p)$ includes time and task penalties; comparisons report both success at matched budgets and the full success–probe-cost tradeoff.

Controlled cases vary posterior entropy, differences between context-optimal strikes, cost gaps, and probe informativeness. One case has distinct probe-outcome distributions but a common optimal strike for every plausible context, isolating Proposition 4. Another has genuinely different optimal strikes and informative outcomes. Within-memory uncertainty is varied separately from context uncertainty to assess the limitation of context-only reweighting. The pure-information diagnostic fixes the post-probe decision state; physical-nudge trials separately account for the state change and compare the approximate selection score with realized total task cost.

Readout and falsification. The probing figure reports gross and net estimated information value, realized cost, success, probe count, and decision latency. In the common-optimum diagnostic, exact value of information is zero even when mutual information is positive; Monte Carlo estimation error is displayed separately. General probing rates are analyzed jointly with posterior masses, cost gaps, and outcome informativeness, rather than assumed to depend only on entropy or action disagreement. Fewer probes at inferior success does not establish a benefit. A failure to improve the success–cost tradeoff, or systematic errors caused by neglected within-memory learning and probe-induced motion, limits the decision-aware approximation.

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